Project - Low Level Design

On

Sports Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Problem Statement & Objectives

Problem Statement

Sports data is highly dynamic, distributed across multiple sources, and difficult to interpret manually in real time. Coaches, analysts, and fans need automated insights such as player performance trends, match outcomes, and statistical comparisons. Existing tools often provide raw statistics but lack intelligent interpretation, predictive insights, and automated reporting.

Project Objectives

- Automate sports data collection from APIs and databases.
- Analyze match and player performance statistically.
- Generate meaningful insights from historical and live match data.
- Provide real-time analytics dashboards.
- Enable predictive modeling for performance and outcomes.
- Deliver automated reports and summaries using AI.

Scope of the Project

- Works with historical, recent, and live match data.
- Supports match-level and player-level analytics.
- Provides statistical insights, trend detection, and comparisons.
- Integrates APIs, databases, and visualization dashboards.
- Can be extended to multiple sports and leagues.

Proposed Solution

Develop an AI-powered sports analytics agent that collects data from sports APIs, processes it using statistical and machine learning techniques, and presents insights through dashboards and automated reports.

The agent performs:

1. Data ingestion from external APIs.

2. Data preprocessing and storage.
3. Analytical computation.
4. Insight generation.
5. Visualization and reporting.

Key Features

- Live match tracking and analytics
- Player performance analysis
- Team comparison engine
- Historical trend analysis
- Predictive performance modeling
- Automated match summary generation
- Interactive dashboards
- API-based data retrieval
- Query-based insight generation

Overall Architecture / Workflow

1. Data Acquisition Layer

- Fetch live, recent, and historical match data using APIs.
- Validate and normalize incoming data.

2. Data Storage Layer

- Structured storage using relational database.
- Tables include:
    - Matches
    - Players
    - Teams
    - Performance statistics

3. Data Processing Layer

- Cleaning and transformation
- Feature engineering
- Statistical calculations

4. Analytics Engine

- Performance metrics calculation
- Trend detection algorithms
- Comparative analytics
- Predictive modeling

5. AI Insight Generator

- Natural language explanations
- Match summaries
- Key performance highlights

6. Visualization Layer

- Dashboards
- Graphs and charts
- Report generation

7. User Interface

- Query input
- Match selection
- Insight display

Tools & Technologies Used

- Python
- Streamlit (UI)
- REST APIs (sports data providers)
- Pandas / NumPy (data processing)

- Scikit-learn (ML models)
- SQL / SQLite / PostgreSQL (database)
- Matplotlib / Plotly (visualization)
- LLM for insight generation

Results & Output

Screenshots / Outputs

- Match statistics dashboard
- Player performance charts
- Team comparison graphs

Reports / Dashboards / Models

- Match summary report
- Player trend report
- Performance prediction model

Key Outcomes

- Automated data retrieval and analysis
- Real-time performance monitoring
- Insightful statistical reporting
- Predictive analytics capability

Conclusion

The Sports Analysis Agent successfully automates sports data collection, processing, and interpretation. It transforms raw match statistics into meaningful insights using AI and analytics. The system improves decision-making for analysts and enhances user understanding through visual and textual reports. The project demonstrates practical implementation of data analytics, machine learning, and intelligent agent architecture.

Future Scope & Enhancements

- Multi-sport support (football, basketball, etc.)
- Deep learning-based prediction models
- Player injury risk analysis
- Fantasy sports recommendation engine
- Voice-enabled analytics assistant
- Cloud deployment for scalability
- Real-time alert system

- ecks for execution results
- Role-based access for administrative operations

These measures ensure robustness and future readiness for real-world deployment.

Performance Aspects

Performance requirements focus on planning efficiency and scalability.

Key performance goals include:

- Low response time for goal interpretation
- Efficient task decomposition and scheduling
- Ability to handle multiple concurrent planning requests
- Predictable execution timelines under load

The modular agent-based design ensures that performance bottlenecks can be isolated and optimized independently.

References

The system design is informed by established research and literature in artificial intelligence and autonomous planning:

- Artificial Intelligence: A Modern Approach, Stuart Russell & Peter Norvig
- Research on intelligent agents and deliberative planning systems

- Academic studies on Agentic AI architectures
- Literature on AI-driven sports analytics and decision-support systems